



Featured Article

Breast cancer trends and outcomes among young Black Women: A national cancer database analysis[☆]Danielle M. Thompson^{a,b}, Kristine Kuchta^{a,c}, Sandra Simovic^a, Katharine A. Yao^{a,b,*}^a Department of Surgery, Endeavor Health, Evanston, IL, USA^b Department of Surgery, University of Chicago, Chicago, IL, USA^c Biostatistical Core, NorthShore University HealthSystem, Evanston, IL, USA

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ABSTRACT

Introduction: This study examined recent trends and outcomes in breast cancer among Black women under age forty.

Methods: Breast cancer incidence data from the National Cancer Database (2010–2022) were analyzed using U.S. Census data population estimates.

Results: Among 188,311 women under forty, 16.4 % were Black, compared to 11.3 % in older Black women. From 2010 to 2022, incidence in young Black women rose 5.3 %, compared to 19.9 % in White, 27.7 % in Hispanic, and 18.9 % in Asian/Pacific Islander women. Triple-negative breast cancer was more common in young Black women (16.3 %) vs. young Non-Black women (12.1 %). Black women under forty had 50 % higher odds of death (OR 1.50; 95 % CI: 1.46–1.55, $p < .0001$) compared to their Non-Black peers after adjusting for age, tumor characteristics, and treatment.

Conclusions: Young Black women have more aggressive breast cancers and worse overall survival than Non-Black young women, though incidence has remained stable over the past decade.

1. Introduction

Breast cancer is the most diagnosed cancer among women worldwide, and studies have shown an increasing incidence of breast cancer in younger women.^{1,2} The definition of “young women” is subject to variability but many clinical trials and other studies have defined it as under 40 years old.^{3,4} Among these younger women, Non-Hispanic Black or African American women face a disproportionately high burden of the disease.^{5,6} This group has not only been more likely to be diagnosed at younger ages compared to other racial and ethnic groups, but also often presents with more aggressive tumor subtypes, such as triple-negative breast cancer (TNBC), which is associated with poorer prognoses and limited treatment options.⁷ Black women, regardless of age, are more likely to be diagnosed at later stages and are 41 % more likely to die from the disease than White women, despite lower incidence rates.⁸

A large recent study by the American Cancer Society showed that breast cancer incidence has been rising steadily, particularly among younger women. Between 2012 and 2021, breast cancer incidence

increased by 1 % annually overall, with a sharper rise of 1.4 % per year in women under fifty. The study also showed that while the incidence rate of Black women was 5 % lower than that of White women, their mortality rate was 38 % higher. Black women were also more likely to be diagnosed at later stages and with more aggressive subtypes, such as triple-negative breast cancer, contributing to lower survival rates across all stages. Additionally, younger Black women had the highest incidence of breast cancer before age 40 compared to other racial and ethnic groups.⁹ Although the American Cancer Society study defined “young-onset breast cancer” as occurring under age fifty, this study uses a cutoff of less than forty years for this analysis in alignment with clinical trials and biologically distinct subgroups. This definition is widely used in breast oncology studies to capture the unique tumor biology and treatment considerations associated with very early-onset disease.^{4,10–13}

To confirm these trends amongst Black women, we examined data from the National Cancer Data Base (NCDB). This database examines cancer cases and treatment from patients treated at approximately 1400 Commission on Cancer accredited facilities across the United States.¹⁴ Studies have shown that the NCDB covers approximately 82 % of breast

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cancer patients across the United States.¹⁵ The primary objective of this study was to analyze trends in the incidence of breast cancer in women under forty years old with a specific focus on Black women. We also examined tumor characteristics and overall survival for Black young women compared to Non-Black young women.

2. Methods

2.1. Study design

This study employed a retrospective cohort design to analyze trends in breast cancer diagnoses among women under and over forty years of age from 2010 to 2022 using data from the NCDB. The NCDB, a nationwide clinical oncology database, collects data from more than 1400 Commission on Cancer-accredited facilities, representing approximately 70 % of all newly diagnosed cancers in the United States.¹⁶ This database includes comprehensive information on patient demographics, tumor characteristics, treatments, and outcomes.¹⁴ The study extracted demographic, clinical, and pathological variables to evaluate trends in breast cancer diagnoses, stratified by age, race/ethnicity, and tumor subtype. To contextualize findings within the broader population, United States Census data were incorporated to standardize rates and calculate population-based denominators for analysis.

2.2. Study setting

The study was set in the United States, using the NCDB to evaluate breast cancer diagnoses across a diverse cohort of women. Data collection spanned from 2010 to 2022, encompassing urban, suburban, and rural healthcare settings. The NCDB's national scope provided a comprehensive overview of breast cancer trends across racial and ethnic groups, offering insights into disparities and variations in diagnosis patterns by age and tumor characteristics.

2.3. Participants

The study included women with newly diagnosed breast cancer of all racial and ethnic backgrounds who were diagnosed with breast cancer between 2010 and 2022. Participants included patients having a combination of treatment and diagnosis at the reporting facility. Participants were stratified into two primary age groups: women under forty years and women over forty years. Annual demographic data from the United States Census from 2010 to 2022 were utilized to represent the total population and calculate standardized incidence rates. Subgroup analyses were conducted to explore trends among Black women, White women, Hispanic women, Asian/Pacific Islander women and women that identified as Non-Hispanic other.

2.4. Data sources

The primary data source for this study was the NCDB, which houses tumor and treatment data on newly diagnosed breast cancer patients treated at Commission on Cancer sites. The database includes variables such as patient age, race/ethnicity, tumor size, stage at diagnosis, receptor status (ER, PR, HER2), and treatment modalities. Additionally, United States Census data were used to standardize breast cancer incidence rates and ensure population-based comparisons. United States Census data were obtained from the United States Census Bureau.¹⁷

2.5. Statistical methods

Descriptive statistics were used to summarize baseline characteristics of the study population, including age, race/ethnicity, tumor subtype, and stage at diagnosis. Incidence rates were standardized using population data from the United States Census. Subgroup analyses examined differences in diagnosis rates, tumor characteristics, and trends by age

group and racial/ethnic categories. Kaplan-Meier analysis and multi-variable Cox proportional hazards modeling were used to assess overall survival between young Black women and Non-Black young women. Statistical significance was determined at a threshold of $p < 0.05$, and all analyses were conducted using statistical software.

3. Results

3.1. Trends in Breast Cancer Incidence in Black Women

Table 1 summarizes the characteristics of female patients diagnosed with breast cancer from 2010 to 2022, stratified by age group (<40 vs. ≥ 40 years). Among patients under forty, the mean age at diagnosis was 34.7 years, compared to 62.6 years in those aged forty or older. The NCDB captured 188,311 breast cancer cases in women under forty years and 4,007,073 cases in women forty and older. Within the younger cohort, 89.1 % were aged 30–39, while 10.9 % were under thirty.

Among breast cancer patients under age forty, 63.5 % were White, 16.4 % Black, 11.2 % Hispanic, 6.3 % Asian/Pacific Islander, and 2.6 % Non-Hispanic Other. In contrast, among patients aged forty and older, White women comprised a larger majority (77.5 %), with lower proportions of Black (11.3 %) and Hispanic (5.6 %) patients.

Table 1
Patient Demographics by Age Group (<40 vs ≥ 40)
Characteristics of all women diagnosed with breast cancer from 2010 to 2022, stratified by age. Includes age distribution, race/ethnicity, insurance type, facility type, and geographic location.

	All Women <40	All Women ≥40
Total Patients	188,311	4,007,073
Age, years [N (%)]		
< 30	20580 (10.9)	0 (0.0)
30–39	167,731 (89.1)	0 (0.0)
40–49	0 (0.0)	672524 (16.8)
50–59	0 (0.0)	1002346 (25.0)
60–69	0 (0.0)	1136101 (28.4)
70–79	0 (0.0)	812142 (20.3)
≥ 80	0 (0.0)	383960 (9.6)
Race/Ethnicity [N (%)]		
Non-Hispanic White	119532 (63.5)	3104836 (77.5)
Non-Hispanic Black or African American	30,808 (16.4)	451145 (11.3)
Hispanic Origin	21162 (11.2)	225817 (5.6)
Non-Hispanic Asian/Pacific Islander	11880 (6.3)	154079 (3.9)
Non-Hispanic Other	4929 (2.6)	71196 (1.8)
Insurance [N (%)]		
Private	141966 (75.4)	1984330 (49.5)
Medicare	3690 (1.9)	1624217 (40.5)
Medicaid	28115 (14.9)	227868 (5.7)
Other Government	3030 (1.6)	40347 (1.0)
Unknown	3994 (2.1)	63390 (1.6)
Uninsured	7516 (3.9)	66921 (1.7)
Facility Type [N (%)]		
CCP	10368 (5.5)	289365 (7.2)
CCCP	67601 (35.9)	1659039 (41.4)
Academic/Research	71662 (38.1)	1188955 (29.7)
INCP	38680 (20.5)	869714 (21.7)
Facility Location [N (%)]		
New England	10520 (5.6)	255358 (6.4)
Middle Atlantic	29369 (15.6)	636348 (15.9)
South Atlantic	41380 (21.9)	887055 (22.1)
East North Central	30047 (15.9)	681870 (17.0)
East South Central	11215 (5.9)	240226 (6.0)
West North Central	13340 (7.1)	292673 (7.3)
West South Central	18548 (9.9)	321355 (8.0)
Mountain	9216 (4.9)	184480 (4.6)
Pacific	24676 (13.1)	507708 (12.7)
^a Mean Age =	34.7	^a Mean Age = 62.6

^a CCP = Community Cancer Program; CCCP = Comprehensive Community Cancer Program; Academic/Research = Includes NCI-designated comprehensive cancer centers; INCP = Integrated Network Cancer Program.

Table 2 presents the annual incidence rates of breast cancer cases per 100,000 population among women aged 18–39 years, stratified by race/ethnicity, between 2010 and 2022. The overall incidence increased from 92 per 100,000 in 2010 to 107 per 100,000 in 2022, representing a 15.4 % relative increase among all races/ethnicities. White and Black populations exhibited an absolute increase of 4 cases per 100,000 and 1 case per 100,000, corresponding to a 19.9 % and 5.3 % relative change, respectively. Among Hispanic individuals, incidence rose from 11 per 100,000 in 2010 to 14 per 100,000 in 2022, reflecting the largest relative increase of 27.7 %. Asian/Pacific Islander populations showed a rise in incidence from 20 to 23 per 100,000 during the same period, a relative change of 18.9 %. The Non-Hispanic Other group demonstrated a slight increase from 16 to 18 per 100,000, equating to a 12.2 % relative change. These findings highlight an overall upward trend in breast cancer incidence rates among young women across all racial/ethnic groups, with the Hispanic population experiencing the greatest relative increase and Black women experiencing the lowest relative increase.

Fig. 1 illustrates the incidence of breast cancer cases per 100,000 population among Black women under and over the age of forty from 2010 to 2022. The incidence among Black women under forty remained relatively stable, ranging from 24 to 26 cases per 100,000 with a total increase of 5.3 % over the study period. In contrast, women aged forty and older experienced a more pronounced increase in incidence, rising from 249 to 299 cases per 100,000, a relative increase of 20.2 %. The disparity between the two age groups persisted across all years, with breast cancer incidence consistently higher among older women.

3.2. Tumor characteristics in young Black Women

Table 3 compares tumor characteristics across four groups: Black women under age forty, Black women aged forty and older, Non-Black women under forty and Non-Black women aged forty and older. Among younger Black women, 63.6 % had hormone receptor-positive (ER+/PR+) tumors, slightly lower than both older Black women (70.8 %) and younger Non-Black women (68.9 %). HER2 positivity was more common among younger Black women (15.1 %) than among older Black women (9.3 %) but comparable to younger Non-Black women (16.4 %). Triple-negative breast cancer was more prevalent in younger Black women (16.3 %) than in both older Black women (12.7 %) and younger Non-Black women (12.1 %).

Younger Black women had a significantly higher proportion of poorly differentiated tumors (56.3 %) compared to older Black women (38.9 %) and younger Non-Black women (49.8 %). Well-differentiated tumors were less common in this group (25.6 %) versus 38.4 % in older Black women and 31.6 % in younger Non-Black women. The proportion of moderately differentiated tumors was also lowest among younger Black women (10.0 %).

Tumor size patterns revealed that younger Black women were more likely to present with larger tumors. Only 29.9 % of this group had tumors ≤2 cm, compared to 47.2 % of older Black women and 36.3 % of younger Non-Black women. Nearly one-fifth (18.8 %) of younger Black

women had tumors >5 cm, nearly double the proportion seen in older Black women (10.6 %).

Lymph node involvement was more frequent among older Black women (55.0 %) than younger Black women (47.4 %) but similar to younger Non-Black women (52.6 %). Stage at diagnosis also varied: 70.9 % of younger Black women were diagnosed at early stage (I/II), compared to 80.2 % of older Black women and 77.0 % of younger Non-Black women. Late-stage diagnoses (III/IV) were more common in younger Black women (25.2 %) compared to the other groups.

3.3. Facility characteristics of young Black Women

Table 4 summarize facility type and geographic location for Black women under age forty, Black women aged forty and older, and Non-Black women under forty. Among Black women under forty, the most common facility type was academic/research institutions (40.9 %), followed by comprehensive community cancer programs (33.6 %), and integrated network programs (20.9 %). This distribution was comparable among Black women aged forty and older. Non-Black women under forty had a slightly lower proportion treated at academic/research centers (37.5 %) and a slightly higher proportion at comprehensive community cancer programs (36.4 %).

Geographically, Black women under forty were most frequently treated in the South Atlantic region (37.9 %), followed by the Middle Atlantic (14.3 %) and East North Central (14.2 %) regions. This regional distribution was consistent across both age groups of Black women. In contrast, Non-Black women under forty were more widely distributed, with higher proportions treated in New England, the Pacific, and Mountain regions compared to Black women.

3.4. Overall survival of young Black Women

Table 5 and **Fig. 2** summarize overall survival among Black women under forty compared to Non-Black women under forty, adjusting for clinical and treatment variables. Black women under forty had lower survival, with a 5-year survival rate of 81.5 % and 10-year survival of 73.0 %, compared to 89.7 % and 82.7 %, respectively, for Non-Black women under forty. Multivariable analysis showed that Black women under forty had 50 % higher odds of death (OR 1.50; 95 % CI: 1.46–1.55, $p < .0001$) than their Non-Black peers after adjusting for age, tumor characteristics, and treatment. To provide additional context, **Table 6** includes 5 and 10 year survival estimates for Black and Non-Black women both under and over age forty, highlighting the persistence of survival disparities across both age and racial groups.

Additional factors associated with improved survival included older age (30–39 vs. <30 years), which may reflect increased likelihood of receiving adjuvant therapies such as chemotherapy, radiation, and hormone therapy, all of which were independently associated with improved survival in our multivariable model. Conversely, higher comorbidity scores, larger tumors, node positivity and receipt of chemotherapy were associated with worse outcomes.

Table 2
Breast Cancer Incidence Trends Among Women Aged 18–39, by Race/Ethnicity (2010–2022)

Annual incidence of breast cancer per 100,000 women aged 18–39, stratified by race/ethnicity. Rates were standardized using U.S. Census data. Hispanic women experienced the highest relative increase over the study period.

Female 18–39 Breast Cancer Cases per 100,000 Population														Change	% Change
Race	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2010–2022	2010–2022
Non-Hispanic White	21	22	22	23	24	24	26	26	26	27	26	27	26	4	19.9 %
Non-Hispanic Black or African American	24	23	25	25	24	26	28	27	27	29	27	28	26	1	5.3 %
Hispanic Origin	11	10	11	12	12	12	13	14	14	15	14	15	14	3	27.7 %
Non-Hispanic Asian/Pacific Islander	20	20	20	20	20	21	21	22	24	25	23	27	23	4	18.9 %
Non-Hispanic Other	16	19	17	18	19	19	20	19	20	21	17	21	18	2	12.2 %
Total	92	93	95	98	99	102	107	108	111	116	107	118	107	14	15.4 %

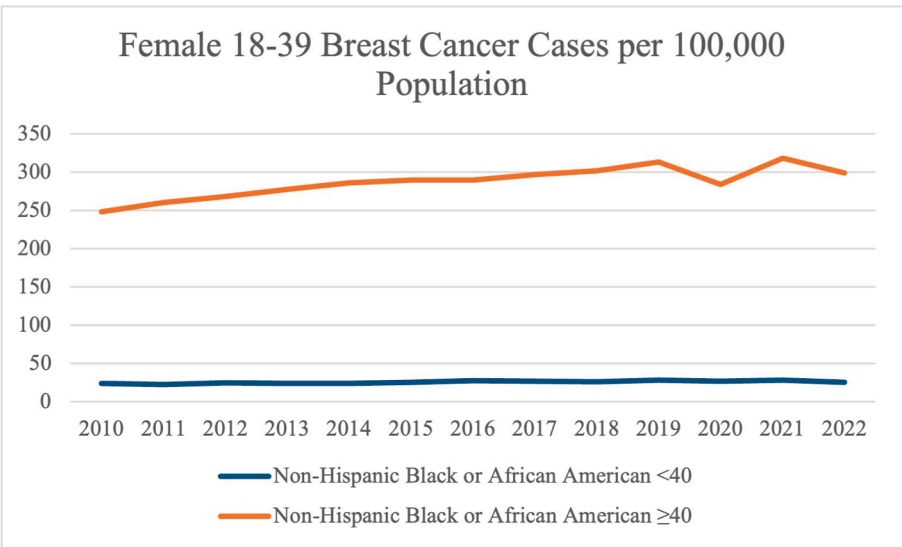


Fig. 1. Breast Cancer Incidence in Black Women <40 vs. ≥ 40 (2010–2022)
Trends in breast cancer incidence among Black women, showing a stable pattern in women under 40 and a 20.2 % increase among women aged 40 and older.

Table 3
Tumor Characteristics by Age and Race
Tumor subtype, grade, size, lymph node involvement, and stage at diagnosis among Black women <40, Black women ≥40, Non-Black women <40, and Non-Black women ≥40. P-values shown for comparisons between each pair.

Tumor Characteristics	Black Women <40 [N (%)]	Non-Black Women <40 [N (%)]	p-value	Black Women ≥40 [N (%)]	Non-Black Women ≥40 [N (%)]	p-value
Total Patients	30,808	157503	<.0001	451,145	3,555,928	<.0001
Hormone Receptor Positive (ER+/PR+)	19603 (63.6)	108488 (68.9)	<.0001	319301 (70.8)	2843268 (80.0)	<.0001
HER2+	4658 (15.1)	25779 (16.4)	<.0001	41890 (9.3)	282873 (8.0)	<.0001
Triple Negative	5020 (16.3)	19043 (12.1)	<.0001	57376 (12.7)	206195 (5.8)	<.0001
Tumor Grade			<.0001			<.0001
Well-differentiated	7893 (25.6)	49722 (31.6)		173257 (38.4)	1765068 (49.6)	
Moderately differentiated	3091 (10.0)	17378 (11.0)		61187 (13.6)	503809 (14.2)	
Poorly differentiated	17349 (56.3)	78361 (49.8)		175265 (38.9)	974827 (27.4)	
Unknown	2475 (8.0)	12042 (7.7)		41436 (9.2)	312224 (8.8)	
Tumor Size			<.0001			<.0001
≤2 cm	9209 (29.9)	57229 (36.3)		213065 (47.2)	2010257 (56.5)	
>2-5 cm	13070 (42.4)	64561 (41.0)		139073 (30.8)	958350 (27.0)	
>5 cm	5804 (18.8)	22712 (14.4)		47908 (10.6)	241073 (6.8)	
Unknown	2725 (8.9)	13001 (8.3)		51099 (11.3)	346248 (9.7)	
Lymph Node Status			<.0001			<.0001
Positive	14709 (47.7)	82807 (52.6)		248130 (55.0)	2165521 (60.9)	
Negative	12160 (39.5)	58269 (37.0)		109992 (24.4)	728558 (20.5)	
Unknown	3939 (12.8)	16427 (10.4)		93023 (20.6)	661849 (18.6)	
Stage at Diagnosis			<.0001			<.0001
Early stage (I/II)	21832 (70.9)	121224 (77.0)		361759 (80.2)	3040214 (85.5)	
Late stage (III/IV)	7769 (25.2)	29950 (19.0)		70099 (15.5)	372981 (10.5)	
Unknown	1207 (3.9)	6329 (4.0)		19287 (4.3)	142733 (4.0)	

4. Discussion

This study provides an updated analysis of breast cancer trends, tumor characteristics, and cancer outcomes in Black women under forty years old using a large, nationally representative dataset. We found that young Black women remain disproportionately affected by triple negative breast cancer, a more aggressive breast cancer phenotype but the incidence of breast cancer among young Black women has remained relatively constant over the past decade. In contrast, the incidence of breast cancer among Hispanic, White and Asian/Pacific Islander young women has increased by 27.7 %, 19.9 %, and 18.9 % respectively. By 2022, incidence rates among young Black and White women had converged at 26 per 100,000, a significant shift from 2010 when young Black women had the highest rates. Despite a relatively modest 5.3 %

increase in incidence among young Black women, their tumor biology, and treatment outcomes continue to reflect deep disparities. Our findings are consistent with recent data from the American Cancer Society, which report that breast cancer incidence is rising most rapidly in younger women, particularly among Hispanic and Asian American populations. The American Cancer Society study reports a 1 % per year increase in the incidence of breast cancer in young women under fifty years old and our study showed a 15 % increasing incidence in breast cancer among young women under forty years old over the past decade. The American Cancer Society also highlights that Black women under forty have the highest incidence of breast cancer compared to all other racial/ethnic groups, but face the highest mortality rates due to later-stage diagnoses and more aggressive subtypes.⁹ While White women continue to have the highest overall incidence, mortality rates

Table 4

Facility Type and Geographic Distribution by Age and Race
Distribution of facility type and geographic region of care for Black women <40, Black women ≥40, and Non-Black women <40 diagnosed with breast cancer.

Facility Characteristics	Black Women <40 [N (%)]	Black Women ≥40 [N (%)]	Non-Black Women <40 [N (%)]
Facility Type [N (%)]			
CCP	1405 (4.6)	23872 (5.2)	8963 (5.7)
CCCP	10352 (33.6)	153806 (34.1)	57249 (36.4)
Academic/Research	12587 (40.9)	176300 (39.1)	59075 (37.5)
INCP	6464 (20.9)	97167 (21.5)	32216 (20.5)
Facility Location [N (%)]			
New England	879 (2.9)	11714 (2.6)	9641 (6.1)
Middle Atlantic	4405 (14.3)	71107 (15.8)	24964 (15.9)
South Atlantic	11690 (37.9)	169058 (37.5)	29690 (18.9)
East North Central	4359 (14.2)	72088 (15.9)	25688 (16.3)
East South Central	2810 (9.1)	39271 (8.7)	8405 (5.3)
West North Central	1247 (4.1)	15690 (3.5)	12093 (7.7)
West South Central	3682 (11.9)	47217 (10.5)	14866 (9.4)
Mountain	317 (1.0)	3855 (0.85)	8899 (5.7)
Pacific	1419 (4.6)	21145 (4.7)	23257 (14.8)

*CCP = Community Cancer Program; CCCP = Comprehensive Community Cancer Program; Academic/Research = Includes NCI-designated comprehensive cancer centers; INCP = Integrated Network Cancer Program.

Table 5

Multivariable Model of Overall Survival for Black vs Non-Black Women <40
Adjusted odds ratios for overall survival among Black vs. Non-Black women <40, controlling for age, tumor biology, stage, node status, and treatment factors. Includes 5- and 10-year survival estimates.

	OR (95 % CI)	P-Value
Age, 30–39 vs. < 30 years	0.85 (0.82–0.88)	<.0001
Income		
\$38,000 - \$62,999 vs. < \$38,000	0.90 (0.87–0.93)	<.0001
≥ \$63,000 vs. < \$38,000	0.77 (0.74–0.81)	<.0001
Education-High School Degree		
80–93 % vs. < 79 %	0.96 (0.93–0.99)	0.0280
> 93 % vs. < 79 %	0.88 (0.84–0.93)	<.0001
Distance from Facility		
11–25 vs. 0–10 miles	1.05 (1.02–1.08)	0.0007
> 25 vs. 0–10 miles	1.04 (1.00–1.07)	0.0350
Charlson Deyo Comorbidity Score		
1–2 vs. 0	1.17 (1.11–1.23)	<.0001
>2 vs. 0	2.03 (1.82–2.27)	<.0001
ER Status, Positive vs. Negative	0.95 (0.91–0.99)	0.0104
PR Status, Positive vs. Negative	0.79 (0.76–0.82)	<.0001
HER2 Status, Positive vs. Negative	0.63 (0.60–0.66)	<.0001
Tumor Size		
1.0–1.9 vs. < 1.0 cm	1.18 (1.11–1.25)	<.0001
2.0–3.9 vs. < 1.0 cm	1.56 (1.48–1.65)	<.0001
≥ 4.0 vs. < 1.0 cm	2.66 (2.51–2.81)	<.0001
Node Status, Positive vs. Negative	2.90 (2.82–2.99)	<.0001
Radiation Therapy, Yes vs. No	0.83 (0.81–0.85)	<.0001
Chemotherapy, Yes vs. No	1.62 (1.56–1.68)	<.0001
Hormone Therapy, Yes vs. No	0.73 (0.71–0.76)	<.0001
Immunotherapy, Yes vs. No	0.85 (0.80–0.89)	<.0001
Race, Black vs. Non-Black	1.42 (1.38–1.46)	<.0001

remain 38 % higher in Black women, a disparity largely attributed to higher rates of triple-negative breast cancer, limited access to early detection, and systemic inequities in care delivery.^{18–20} Likewise, our study shows that young Black women have worse overall survival than young Non-Black women with a 50 % worse overall mortality despite adjusting for tumor characteristics and treatments. The NCDB does not contain any recurrence information, so we do not know if this increased mortality is related to breast cancer or not. However, given the young

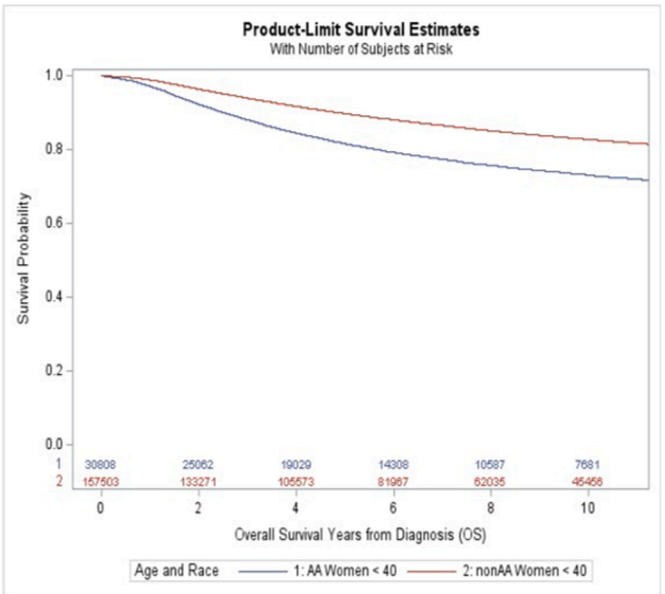


Fig. 2. Kaplan-Meier Overall Survival Curve for Black vs. Non-Black Women <40

Overall survival from time of diagnosis, comparing Black women under age 40 to Non-Black women under age 40. Black women had significantly lower 5- and 10-year survival rates.

Table 6

Five- and Ten-Tear Overall Survival Estimates Stratified by Age and Race.
Young Black women (<40) had the lowest 10-year survival compared to all other groups. Disparities persisted across both age and racial groups.

	Black Women <40	Non-Black Women <40	Black Women ≥40	Non-Black Women ≥40
5 Year Survival	81.53 %	89.73 %	80.68 %	86.49 %
10 Year Survival	73.00 %	82.69 %	67.78 %	76.89 %

age of these women and the lower likelihood of other life-threatening conditions or co-morbidities, it is more likely these deaths were from breast cancer. Nonetheless, these data are very bothersome and certainly warrant further investigation.

The persistently high prevalence of triple-negative breast cancer (16.3 %) and poorly differentiated tumors (56.3 %) in young Black women in our cohort underscores the unique biologic profile of breast cancer in this population. Compared to both older Black women and young Non-Black women, those under forty were more likely to present with larger tumors, advanced stage disease, and lymph node involvement, all factors associated with poor prognosis. These findings are supported by previous research demonstrating that African ancestry are independently associated with more aggressive tumor phenotypes, including basal-like and HER2-enriched subtypes.^{21,22} Our findings also align with studies which have demonstrated that very young age at diagnosis is independently associated with more adverse tumor biology and poorer survival outcomes.^{23–25} These more aggressive tumor features also likely contribute to the worse overall survival we are seeing in these young Black women.

The sharp rise in breast cancer incidence among young Hispanic and Asian women is notable and warrants further investigation. Possible explanations include population changes in the United States, changes in reproductive behavior, obesity, hormone exposure, acculturation-related lifestyle shifts, and improved access to screening and diagnostic services in these populations. However, the relative contribution of these factors remains unclear. Importantly, the rise in incidence does

not appear to be paralleled by a rise in mortality in these groups, suggesting more favorable tumor biology or better treatment access compared to young Black women.

Facility characteristics and access to care also contributed to disparities observed in our study. Younger women, particularly those identifying as Black, were more likely to be treated at academic or research institutions yet still faced higher odds of poor outcomes compared to their Non-Black peers. Our survival analysis showed that Black women under forty had 50 % higher adjusted odds of death compared to Non-Black women under forty. This finding suggests that factors beyond tumor biology such as delayed care, barriers to timely treatment, financial hardship, and structural racism may contribute to worse outcomes. While our analysis adjusted for area-based measures of socioeconomic status including median income, education level, and travel distance to the treating facility, Black women under forty continued to experience significantly worse survival outcomes, suggesting that structural barriers beyond these factors may contribute to persistent disparities.

Although our study leverages the strength of the NCDB, which captures approximately 80% of all newly diagnosed breast cancers in the United States,¹⁵ several limitations should be noted. The NCDB does not capture patients treated at non-CoC facilities or outside of hospital settings, and lacks granular sociodemographic variables such as income, education, and social determinants of health. Additionally, the NCDB does not capture patient-level data on family history or genetic testing, which limits our ability to evaluate hereditary cancer syndromes or BRCA-associated risk in this population. The retrospective nature of the study precludes causal inference however we have shown similar findings to other studies of young Black women. Despite these limitations, our findings show some significant trends and worrisome outcomes that warrant future investigations to determine causes. These findings suggest actionable targets for clinical and public health interventions.

5. Conclusion

Our study demonstrates that while Black women remain at high risk for early-onset breast cancer, the most pronounced relative increases in incidence are emerging among Hispanic women under forty. These findings highlight the need for targeted public health strategies, improved access to early detection and treatment, and further research into the biological and systemic factors contributing to these disparities. Addressing these challenges are critical to reducing the burden of breast cancer in young women and ensuring equitable care across all racial and ethnic groups.

CRedit authorship contribution statement

Danielle M. Thompson: Methodology, Investigation, Writing – original draft, Conceptualization, Writing – review & editing, Data curation, Formal analysis. **Kristine Kuchta:** Methodology, Formal analysis, Data curation. **Sandra Simovic:** Project administration. **Katharine A. Yao:** Resources, Funding acquisition, Writing – review & editing, Project administration, Conceptualization, Supervision, Investigation, Writing – original draft, Methodology.

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Declaration of competing interest

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